CMZ · PHASE 2 · OCTOBER 2026

Three tasks, one animal, *one pipeline.*

Short-clip behavior classification, detection and silhouette features for a zoo civet, plus an activity table for a night of banded civet.



PROJECT TEAM, CMU

CAMERA 02 · 19 MAY 2026

A keeper cannot watch every hour.

WHY THE PROJECT EXISTS

01

Four rooms. One animal. Four cameras.

The project setting from the brief: a civet enclosure observed by fixed cameras, day and night. Four rooms belong to one proposed monitoring context.

4	4	1
ROOMS	CAMERAS	FOCAL ANIMAL

> A design setting, not verified coverage. The clips we received do not map to specific rooms.



Welfare is read off behavior.

Three things a keeper watches for, and each one is a duration, not a snapshot.

01	**Is it eating?**	Food-directed intake at the bowl. **A drop in feeding time is an early signal.**
02	**Is it moving?**	Travel around the enclosure. **Lethargy shows up as fewer and shorter bouts.**
03	**Is it resting normally?**	Sustained inactivity. **Rest that crowds out the other two is also a signal.**

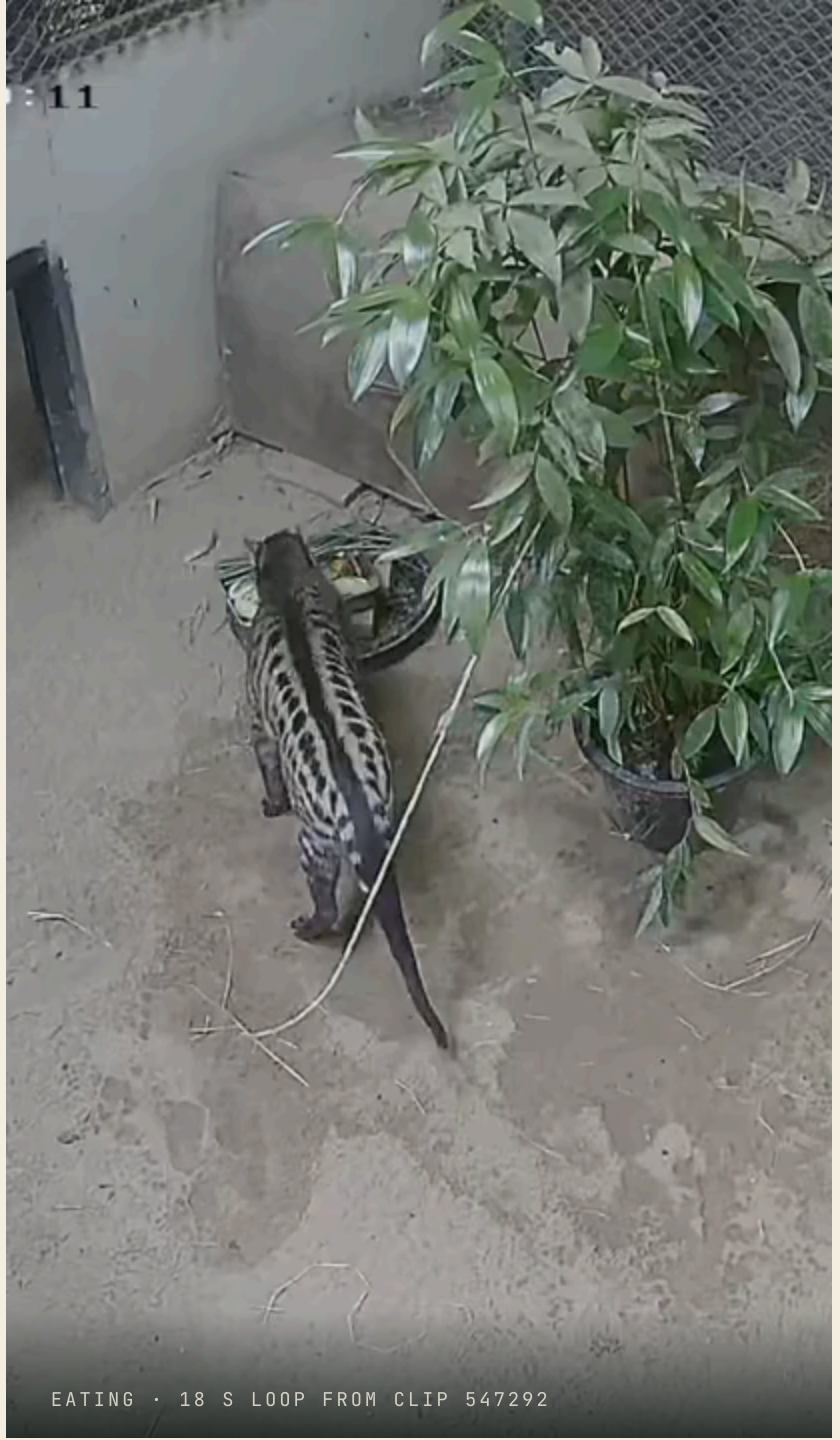
> Premise of this project: the comparison is the animal's own baseline, not a population average.

Our behavior dataset is 17.5 minutes.

A four-camera installation produces 96 hours a day. Hand review does not scale past a demonstration.

22	1,052	26,295	0.3%
SOURCE CLIPS	SECONDS OF FOOTAGE	FRAMES AT 25 FPS	OF ONE DAY, FOUR CAMERAS

> 17.5 minutes is watchable. Seven days of four cameras is 672 hours.



Can the machine do the looking?

Three tasks from the brief, on the same footage, ending in one behavior timeline.

TASK 2 · DETECTION

See

FIND AND BOX THE CIVET IN EVERY SAMPLED FRAME

TASK 1 · CLASSIFICATION

Name

LABEL EACH FIVE-SECOND WINDOW EATING, LOCOMOTION OR RESTING

TASK 3 · ANALYSIS

Measure

BODY ANGLE, HEAD ANGLE, SPEED AND LIMB EXTENT FROM A SILHOUETTE

> Same windows, same timecodes. That is the whole design idea, and the rest of the talk tests whether it holds.

What the brief asked. What we built.

TWO SUBSTITUTIONS, BOTH RECORDED

02

Three tasks, two substitutions.

TASK	BRIEF PATH	WHAT WE BUILT	MEASURED
Task 1 Classification	Walking · Eating Resting · Climbing	Five-second windows, three classes, one frame CNN over an ImageNet backbone	Macro F1 0.949 15 held-out windows
Task 2 Detection	Video → frames → YOLO labels → train	Labels from a per-clip background model. No hand-drawn box at any point	mAP50 0.537 animal found in 59% of frames
Task 3 Analysis	Pose → keypoints → four features	Silhouette geometry with the same four feature names	Macro F1 0.422 locked test windows

> Climbing dropped with instructor confirmation (ADR 0007). Task 3 is a silhouette proxy, not keypoints (ADR 0009).

Behavior, window by window.

THE EVIDENCE BASE

03

138 labelled windows is the whole evidence base.

22	201	138	93 · 30 · 15
SOURCE CLIPS	FIVE-SECOND WINDOWS	LABELLED WINDOWS	TRAIN · VAL · TEST

01 **Verified intervals.** One model class covers the whole span, so a window is kept only when its label is unambiguous.

02 **Dropped windows.** 63 of 201: 46 not covered completely by one interval, 17 touching a Transition or Uncertain span.

03 **Split by whole clip.** No frame from a test clip ever appears in training.

The test set was locked before it was read.

Eight clips are the gold set and also the test split. Two people labelled them blind to each other, then reconciled.

PASS A

Blind

FIRST ANNOTATOR, EIGHT GOLD CLIPS

PASS B

Blind

SECOND ANNOTATOR, SAME CLIPS, NO SHARED ANSWERS

20 OCT

0.597

INTERVAL DICE BETWEEN THE PASSES, THEN RECONCILED

21 OCT

Once

SIGNED, LOCKED, DECODED AND SCORED

> Disagreement was reconciled into one interval set. The test is scored once, so no decision downstream of it could shape the model.

One error in fifteen held-out windows.

0.949

MACRO F1 · 15 HELD-OUT WINDOWS

0 abstentions. Accuracy over decided windows **0.933**.

1 miss: an Eating window predicted Resting at 0.847.

CONFUSION MATRIX · ROWS TRUE, COLUMNS PREDICTED

	EATING	LOCOMOTION	RESTING	TOTAL
Eating · 7	**6**	0	**1**	7
Locomotion · 2	0	**2**	0	2
Resting · 6	0	0	**6**	6
PREDICTED	6	2	7	15

One window, two plausible explanations.

CLIP 25690519193808204

Window 40–45 s

LABELLED Eating → PREDICTED Resting 0.847

> The label was not changed after the test was read.

A model miss

The animal is partly hidden, and a still head over a bowl looks like rest.

An annotation error

The boundary sits close to the edge of the interval, and five seconds is a coarse window.

With one read of the test set, we cannot separate the two.

What 0.949 does not establish.

01	**Fifteen windows**	Locomotion has 2, and all 6 Resting windows come from one clip. A per-class score on that support is a baseline, not a stable estimate.
02	**Camera and class travel together**	Every Resting window is Camera 03 and nearly all Eating is Camera 02. **A score can come from the recording rather than the behavior.**
03	**It is an upper bound**	0.949 describes these clips on these days. It is not an estimate of a new recording.

> Five clips and one camera per class is enough to show the pipeline works, not enough to claim it generalizes.

Nobody drew a box.

YOLO WITHOUT ANNOTATION LABOUR

04

The obvious detectors did not find the civet.

×	`Pretrained YOLO11n` `COCO classes, full frame`	Zero detections of the animal. One potted plant at 0.344.
×	`Frame differencing` `largest moving blob`	Locks onto swaying foliage, which moves more than a resting civet.
×	`Grounding DINO` `open vocabulary, “a civet”`	40+ overlapping boxes, top score 0.535 on empty floor.
✓	`Per-clip median background` `with a persistence filter`	Keeps the animal and drops the foliage. **408 candidate boxes.**

The background model draws the boxes.

Median frame per clip, a per-pixel noise threshold, a component in an animal-size band, and a box must persist for about a second.

01 · SAMPLE
1,061
FRAMES AT 1 FPS ACROSS 22 CLIPS

02 · PROPOSE
470
CANDIDATE BOXES FROM THE BACKGROUND MODEL

03 · FILTER
408
KEPT AFTER THE ONE-SECOND PERSISTENCE RULE

04 · AUDIT
116
BOXES REVIEWED ON CONTACT SHEETS

05 · CHECK
113
WERE ON THE ANIMAL

> The 3 misses were empty floor, all in the last twelve seconds of one clip after the animal left the view.

The detector learns from its own labels.

0.537

MAP50 · ROUND 1 ON SEED LABELS

0.499

MAP50 AFTER SELF-TRAINING, UNTOUCHED VALIDATION

59% of held-out frames fire at confidence 0.25. 138 / 235

5 of 10 audited held-out boxes were on the animal.



5 ON THE ANIMAL · 5 NOT

> Round 2 scores 0.774 on its own validation, but 115 of those labels were partly written by round 1, so that number is not comparable.



YOLO BOX 0.66 · CAMERA 03 · 19 MAY 2026

Four numbers from a shape.

NO KEYPOINTS, NO POSE MODEL

Keypoints became a silhouette.

MEDIAN PER CLASS · TRAINING AND VALIDATION WINDOWS WITH A SILHOUETTE

FEATURE	EATING	LOCOMOTION	RESTING
Body angle	115°	88°	154°
Head angle	113°	49°	96°
Movement speed	17 px/s	134 px/s	2 px/s
Limb extent	56 px	148 px	98 px

A silhouette cannot tell a head from a tail, and a still animal joins the background.

0.479

VALIDATION MACRO F1

0.422

TEST MACRO F1

> Resting scores **0** on test: 58 of 93 training windows have no silhouette at all.

> Per-class F1: Eating 0.267 · Locomotion 1.000 · Resting 0.000. Locomotion separates by an order of magnitude in speed.

One clip, three tasks, one command.

```
$ python3 scripts/demo_all.py data/raw/Eating/25690519193422801.mp4

data/raw/Eating/25690519193422801.mp4  (50.0 s, 10 windows, abstain below 0.00)
  start     end  T1           conf  T2 box              via     conf  crop         body  head   speed  limb
   0.00    5.00  Eating       0.73  408,462,606,528     motion  0.00  Eating      6 161       1    63
   5.00   10.00  Eating       0.69  606,510,674,564     motion  0.37  Eating      25 16      23    52
  10.00   15.00  Eating       0.69  632,472,700,550     motion  0.00  Eating      43 24      20    68
  15.00   20.00  Eating       0.67  341,365,624,534     yolo    0.58  Locomotion  119 19      24    51

detection: 10/10 windows found the animal (6 by the trained detector, 4 by the per-clip model)
full-frame and crop labels agree on 4/10 windows
```

Validation clip. Output trimmed to the header, four data rows and the two summary lines.

One animal, one timeline.

INPUT

FOOTAGE
Video + conditions
`clip · time · light · view`

GRID
Five-second windows

THREE TASKS, THE SAME WINDOWS

TASK 1 · CLASSIFY
Behavior per window, with a confidence

TASK 2 · DETECT
Box per frame, with a per-clip fallback when the detector is unsure

TASK 3 · MEASURE
Body angle, head angle, speed, limb extent

OUTPUT

EVIDENCE
Timeline + summaries
`bouts · duration · counts`

RECORD
clip · start · end · behavior · confidence · conditions

> Every row keeps clip, start, end, behavior, confidence and condition flags, so any summary can be traced back to a timecode.

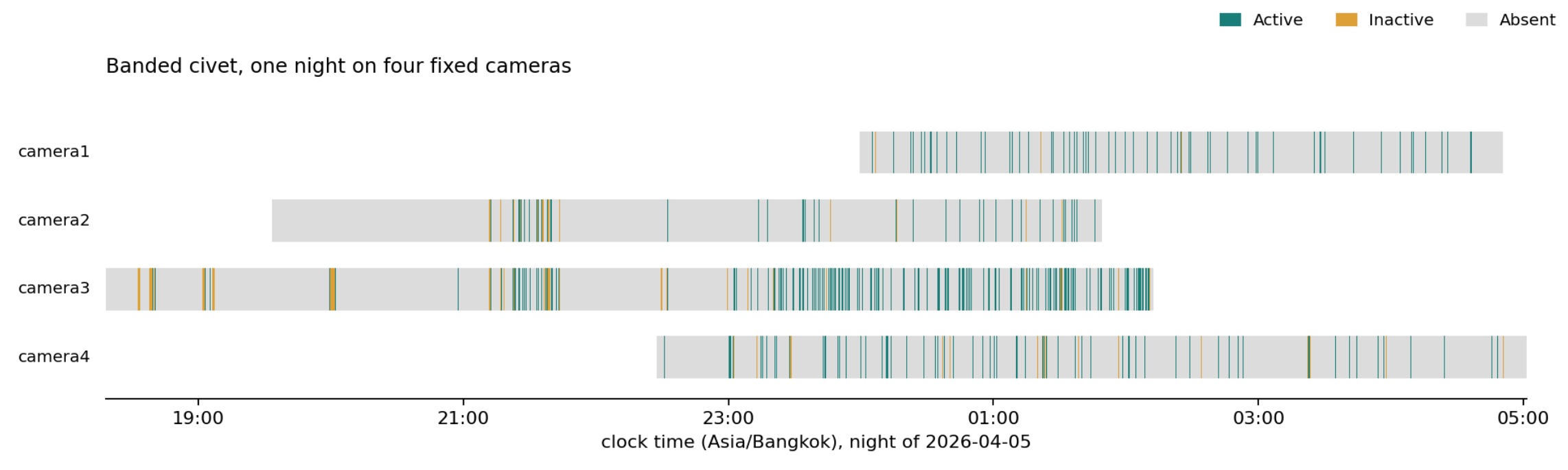
Another species. Another night. No annotation.

WHAT A FIXED CAMERA MEASURES BY ITSELF

06

25.59 hours, and not one label.

Four fixed cameras watched one enclosure on the night of 2026-04-05. The scan measures what the camera can see by itself, which is presence and movement.



Each row is one camera on the same clock. The empty room is the per-pixel median of the whole clip, so a box survives only when the next second corroborates it and an insect appears once while a civet persists.

The table, scored on a camera it never saw.

ACTIVE AND INACTIVE TABLE · SHARE OF EACH CAMERA'S RECORDED TIME

CAMERA	PRESENT	ACTIVE	INACTIVE	ABSENT
camera1 · 4.86 h	8.7%	8.4%	0.4%	91.3%
camera2 · 6.26 h	4.6%	3.3%	1.4%	95.4%
camera3 · 7.90 h	14.6%	11.2%	3.4%	82.3%
camera4 · 6.56 h	8.1%	6.7%	1.4%	91.9%
All · 25.59 h	9.4%	7.6%		

924 active bouts, median 5 s and longest 40 s. The animal crosses a view rather than settling in it.

0.664

MACRO F1 · CAMERA 4, HELD OUT

0.453

ALWAYS ANSWERING ACTIVE

Active precision **0.921**, recall **0.770**.

Inactive precision **0.381**, recall **0.682**.

Validation on camera 3: **0.804**.

> The audit read 24 windows of each state at full resolution: 24 of 24 Active, 23 of 24 Inactive and 22 of 24 Absent held what the label said.

What this does not prove.

01	**Fifteen test windows from five clips**	Locomotion has 2 and Resting takes 6 from one clip. A per-class score on that support is a baseline.
02	**Camera and class travel together**	A score can come from the recording rather than the behavior.
03	**The detector does not generalize**	It fails on the 2026-05-20 recording day, absent from training, and fires on a bright plant.
04	**Task 3 is a proxy**	No keypoints and no pose model. A still animal joins the median background, so most Resting windows have no features.
05	**One training seed**	Seed 0 only. No run-to-run spread, so the split between score and seed is unknown.
06	**One night of banded civet**	25.59 hours from four cameras, with no human label and no second night. The activity score checks the measurement, not a person.

And no human ground truth for detection: its quality is a counted visual audit, not mAP against truth.

Four changes that turn numbers into evidence.

01	**Hold out a recording group**	Collect footage from sessions absent from training and repeat the evaluation, so the upper bound becomes an estimate.
02	**Widen per-class support**	Add Locomotion and Resting test clips from more than one clip per class before reading per-class differences.
03	**Calibrate abstention**	Choose the abstain threshold on a larger validation set or a separate calibration set, never on the test windows.
04	**Revisit Climbing and pose**	Only if verified footage and keypoint labels exist. Until then, three classes and silhouette features are the scope.

CMZ · PHASE 2 · OCTOBER 2026

A reproducible baseline that states what it measured and *what it did not.*

QUESTIONS PROJECT TEAM, CMU

CODE AND REPORTS: CIVET-BEHAVIOR-DETECTION